SLIDE 1: EDGEGUARD-AI - VERIFIABLE EDGE AI & ZERO-TRUST TELEMETRY

Startup Title: EdgeGuard-AI

Founder & Lead Systems Architect: Abdullah TAY (abdullahtay@gmail.com | +90 532 206 61 68)

Repository: https://github.com/abdullahtay/edgeguard-ai (MIT Open-Source)

Target Program: INiTS SCALEup Incubator & STARTKapital / aws Preseed Deep Tech (Vienna, Austria)

Key Technologies: Artificial Intelligence & Data Innovation | Chips, Electronic Components & Embedded Systems

SLIDE 2: THE PROBLEM - UNVERIFIED EDGE TELEMETRY & EU AI ACT COMPLIANC

1. Critical infrastructure, industrial IoT (IIoT), and satellite/GNSS edge nodes suffer from telemetry spoofing, adversarial sensor injection, and silent model drift.
2. Existing cloud-centric security pipelines introduce 50-200ms network latency and violate strict data-sovereignty and bandwidth constraints on embedded ARM64/RISC-V microcontrollers.
3. The EU AI Act (Article 15) and NIS2 Directive legally mandate verifiable robustness, cybersecurity, and tamper-evident audit logs for high-risk AI systems deployed in European industry and mobility.

SLIDE 3: THE SOLUTION & CORE TECHNOLOGY (TRL 4-5 PROTOTYPE)

EdgeGuard-AI is an ultra-low-latency (<50 microseconds) embedded Rust/C++ runtime and zero-trust cryptographic verification engine for edge AI and industrial/satellite IoT sensors:

- Sub-50us Streaming Z-Score & Isolation Forest Anomaly Detection directly on raw sensor frames.
- Zero-Heap Deterministic Ring-Buffer Execution (<256 KB RAM footprint) on ARM Cortex-M/A and RISC-V.
- Hardware-Anchored Ed25519 + SHA-256 Merkle Audit Envelopes proving sensor integrity from chip to cloud.
- Automated EU AI Act Article 15 & IEEE P3195 compliance telemetry generation.

SLIDE 4: UNIQUE SELLING PROPOSITION (USP) & COMPETITIVE ADVANTAGE

- 100x Lower Latency: <50us per frame vs. 5-50ms for Python/container-based edge monitors.
- Cryptographic Provenance: Combines real-time statistical anomaly scoring with zero-trust Ed25519 signing.
- Hardware Agnostic: Runs bare-metal or Linux on STM32, NXP i.MX, NVIDIA Jetson, and RISC-V.
- Open-Core Defensibility: MIT-licensed core verification spec + commercial enterprise fleet attestation SaaS.

SLIDE 5: MARKET OPPORTUNITY & BUSINESS MODEL (B2B DEEP TECH)

- TAM: EUR 18.4B Global Edge AI & Industrial Cybersecurity Market (24.6% CAGR through 2030).
- SAM: EUR 3.2B DACH & European Industrial IoT, Smart Grid, Railway/Mobility & Satellite Telemetry Security.
- SOM (36 Months from Vienna HQ): EUR 15M ARR across 120 European industrial & critical infrastructure custom
- Business Model: Open-Core + B2B Per-Node Annual Licensing (EUR 12-45/node/year) + Enterprise Compliance Control Plane & EU AI Act Audit Certification Dashboard (EUR 25,000-75,000/year per enterprise).

SLIDE 6: TRACTION, VIENNA INCORPORATION & INiTS SCALEUP ROADMAP

- Current Traction: Working open-source prototype (TRL 5) at github.com/abdullahtay/edgeguard-ai with automated CI benchmarks demonstrating 38us median latency and zero-trust Merkle verification.
- Vienna Incorporation Plan: Incorporating EdgeGuard AI GmbH in Vienna during the INiTS 100-Day Startup Camp.
- Use of INiTS STARTKapital (EUR 100k) & aws Preseed (EUR 200k+): Hardware-in-the-loop (HIL) certification with Austrian industrial partners (Siemens Mobility Austria, TTTech, Infineon Austria) & first 5 B2B pilots.